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ML — TP 5: Neural Networks

1. Problem 1

Build a neural network model to classify images of handwritten digits. The dataset can be found in 'tiny_mnist.zip'. There are 3 classes: 0, 1, and 2.

2. Problem 2

Given the complete MNIST dataset (can be found in " sample_data " of Google Colab), build another neural network. There are now 10 classes.

Hint: try to train the network using a small batch from the whole training set for each iteration.

3. Problem 3

Given a dataset about chemical measurements of wine, build a neural network as a classifier to predict the quality (Medium, Good, or Excellent) of wine.
Divide the dataset into 3 subsets: training (60 %), validation (10 %), and test (30 %).
Use the validation set to choose the number of hidden layers (between 2 and 5 hidden layers with 50 units each) for your neural network.
Finally compute the error rate of the model that you select on the test set.

4. Problem 4

From the Khmer character dataset, training and test sets are already provided.
Divide the training set into two subsets: new training set (80 %) and validation set (20 %). Use the validation set to find the best possible neural network model and evaluate it on the test set.

5. Problem 5: Mini Project no. 4

Taking at least 200 photos of two similar animals (example cat and dog) and train a Neural Network model to classify the two animals. Present your experiments to the class next week. Note: the performance of each model should be evaluated on the test set and the test set must not be used during the training process.

Submitted by Mr. TOUCH Sopheak, Mr. Pen Chentra, Ms. Un Sreyvi on 11 de mayo de 2026.